

SUMMARY

Results-driven Senior Cloud DevOps/Site Reliability Engineer with over 6 years building and managing resilient cloud platforms across AWS and Azure. Proficient in IaC, Kubernetes, CI/CD, automation, reliability, security, and observability. Possesses unique experience in architecting and implementing infrastructure for Generative AI and LLM-based applications. Successfully delivered impactful outcomes, including 90% infrastructure provisioning acceleration and 99.8% deployment reliability within finance and e-commerce sectors. Eager to apply advanced Cloud/DevOps practices and AI infrastructure knowledge in a dynamic technical environment.

EXPERIENCE

Sr. DevOps/Site Reliability Engineer | Lenovo, Morrisville, NC | March 2023 - Present

- **Cloud Architecture & Infrastructure:** Led the architecture and implementation of scalable, highly available, multi-tier **Azure** infrastructure supporting global platforms like lenovo.com and AI initiatives (studioai.lenovo.com, smartsales.lenovo.com/chat), designing and deploying network and security controls (**Akamai CDN, Azure Firewall, Azure Content Filter, Key Vault, Managed Identities**) tailored for AKS clusters. Simultaneously, built scalable ETL pipelines on Azure using **Azure Data Factory, Synapse Analytics**, and **Databricks**, implementing **Medallion architecture** for data transformation (from sources like **MongoDB, MySQL, flat files**), which reduced data processing time by 27% and enabled real-time analytics capabilities.
- **Infrastructure as Code (IaC) & Automation:** Designed and implemented a standardized **Terraform/Terragrunt IaC** framework supporting Azure Infrastructure, creating custom modules to ensure reusable and consistent infrastructure provisioning. Integrated this framework with secure GitLab CI pipelines (**Azure SPN, Blob state**) enabling accelerated delivery (by **90%**) and improved consistency for a team of **~40 Ops**. Applied this framework across critical platforms such as lenovo.com, studioai.lenovo.com, and various AI/RAG chatbot initiatives.
- **Kubernetes & Container Orchestration:** Configured and managed 8 high-traffic Azure Kubernetes Service (**AKS**) clusters (**~25 nodes each**) across multiple environments, including lenovo.com. Actively maintained **cluster health** and performed regular **version upgrades** to ensure stability and security. Enhanced cluster traffic control and security by implementing complex routing configurations using **Kubernetes Services, Istio Virtual Services**, and **APISIX** routes to optimize internal **microservice** communication and external API access. Enforced network policies with **Azure CNI** and integrated **Azure policies** for cluster protection and centralized governance of hosted **microservices** and middleware (**Apisix, Zookeeper, Kafka**).
- **CI/CD & Deployment:** Engineered and Operated event-driven **Argo pipelines (Workflows+Sensors)** triggered by GitLab/Kafka **EventSources**, deploying **50+ microservices** via **Argo CD ApplicationSets** and **HelmCharts** with **Harbor** registries, **MinIO** storage integration across AKS clusters, achieving **99.8%** deployment success rate and reducing rollout time by **35%**.
- **AI/LLM Infrastructure:**
 - Led infrastructure design and implementation on **Azure/AKS** to integrate **Generative AI services and Large Language Models (LLMs)** (**GPT, LangChain/Langgraph**) for the internal **StudioAI** platform (studioai.lenovo.com), supporting **~400** internal users.
 - Architected and implemented scalable **Azure solutions** for **RAG-based AI chatbots (SmartSales, Navigator)**, serving thousands of users across **EMEA** & expanding globally on all lenovo.com sites, integrating **Azure OpenAI** and vector search (**Milvus/OpenAI embeddings**).

- Configured Azure OpenAI resource and deployed models (**LLM, text embeddings**) within the **Azure AI foundry**, managing **API access** via secure **secret** injection into **Kubernetes deployments** via **volumes** using the **Azure Key Vault CSI driver** and **Secret Provider Class**.
 - Led **Proof-of-Concept (POC/POT)** initiatives for **Langchain/Langgraph** python based **agentic frameworks**, defining architectural patterns (**edges, llmchain, nodes, tools, memory, state, Fastapi**) and leveraging **Pydantic**, resulting in the adoption of this framework by development teams.
- **System Reliability & Incident Management:** Managed **production** support and **incident response** for the global **Lenovo E-Commerce** application (lenovo.com) across **20+** countries. Resolved critical incidents within **SLA** using **OpenSearch, Splunk APM, Grafana**, conducting **RCA** and **postmortems** for **P2+** events to improve reliability.
- **Observability & Monitoring:** Established and maintained comprehensive observability across platforms including lenovo.com, studioai.lenovo.com, and AI chatbots. Leveraged a diverse stack of tools including **Prometheus, Grafana, ELK (Elasticsearch, Logstash, Kibana), Splunk APM**, and **Percona** for database monitoring, alongside implementing specialized LLM monitoring solutions like **Langfuse** and **Langsmith** to ensure visibility, performance analysis, and proactive issue detection.
- **Database Management:** Managed and optimized performance for critical databases including **PostgreSQL, Redis, MySQL, Mongo**, and **Cosmos DB** supporting platforms like lenovo.com and studioai.lenovo.com. Implemented **MySQL Master-Slave** replication with **Redis** caching to significantly reduce query latency by **50%** and decrease read load on **replicas**, ensuring high performance and reliability while handling operations like **provisioning, user management**, and **schema modifications**.

Site Reliability Engineer (Contractor) | GoldmanSachs(via Origin Hubs), Dallas, TX | April 2022 – Feb 2023

- **Kubernetes & Container Orchestration:** Managed and maintained **11 enterprise-scale Red Hat OpenShift (OCP)** clusters (~20 nodes each) across multiple environments, including **performing cluster version upgrades** and **health maintenance**. Supported application teams by troubleshooting and resolving deployment issues as a Site Reliability Engineer. Integrated **Istio service mesh** (metrics, logs, traces), **VMware Contour (Envoy)** for Ingress, **Cert-Manager** for TLS/SSL certificates, and **Ceph** for storage. Configured scaling (**VPA/HPA**) and managed workloads via **Helm charts** and **Kubernetes YAML manifests**.
- **CI/CD & Deployment Strategies:** Supported **GitOps-based CI/CD** pipeline using **GitLab CI** and **Argo CD** for **enterprise microservices** on **OCP**. Assisted application teams with their **deployment pipelines**, troubleshooting issues related to **Application Sets, sync policies** (automated sync, self-healing, pruning), and advanced deployment strategies like **Blue-Green** and **Canary rollouts** with **Argo Rollouts**, ensuring reliable application delivery.
- **Security & Governance:** Assisted application teams with **security** and **governance** concerns on **OCP** clusters. Contributed to writing **Open Policy Agent (OPA) policies** and troubleshooting deployment failures caused by policy violations. Provided support for secret management workflows utilizing **HashiCorp Vault** and secure certificate issuance using **Cert-Manager** integrated with the **Goldman Sachs PKI**.
- **Observability & Monitoring:** Monitored the health and performance of enterprise-scale **OCP clusters** by leveraging **Prometheus Alert manager** and **Grafana** dashboards (using **PromQL**) and responding to **PagerDuty** alerts. Utilized the observability data collected via **Istio service mesh** to effectively troubleshoot issues and maintain **cluster uptime**, contributing to meeting firm-wide **SLO/SLAs**.
- **System Reliability & Support:** Maintained firm-wide **SLO** and **SLAs** for **Kubernetes clusters**. Provided **L2 support**, responded to outages, and assisted application teams with their **deployments** and troubleshooting on **clusters**.

Cloud/DevOps Engineer (Contractor) | BMO (via Origin Hubs), Chicago, Illinois | Feb 2021 – Mar 2022

- **Cloud Infrastructure (AWS & Azure):** Configured and managed core infrastructure services across **AWS** (VPC, EC2, S3, RDS/Aurora, DynamoDB, IAM, Route53, etc.) and **Azure** (VMs, Virtual Networks, Load Balancers, Storage Accounts etc), focusing on implementing standard **two-tier** and **three-tier** architectures.
- **Infrastructure as Code (IaC) & Automation:** Delivered Infrastructure as Code (**IaC**) solutions across **AWS** and **Azure** using **AWS CDK** (Typescript/Python) and **terraform**. For the [MRNG](#) Project, co-developed **Azure Bicep** scripts with Microsoft personnel for **event-driven** spot **compute** provisioning, alongside automating domain joining via **PowerShell** bootstrap scripts securing credentials from **Key Vault**.
- **Automation:** Automated the installation of security agents (Carbon Black, IBM BigFix) on Windows/Linux VMs using **Ansible Tower** workflows triggered during **EC2 provisioning**.
- **Cloud Migration:** Responsible for the end-to-end migration of **on-premises** systems to **AWS** cloud, which involved designing and implementing the necessary target **cloud infrastructure** on **AWS**. Performed data migrations using **AWS DMS** and transferred **application assets** via AWS SFTP (Transfer Family).

IT Consultant | Villanova University, Villanova, PA | August 2019 – Jan 2021

- **System Management & Security:** Managed **macOS** automation (Jamf), user security (**Active Directory**, **Group Policies**), and network security (**Citrix ADC dual-factor auth**).
- **Monitoring:** Monitored virtual machines using **PRTG** network monitoring.

DevOps Engineer | Aurex Inc, Ashburn, VA | September 2018 – Jul 2019

Built and configured the core cloud infrastructure on **AWS** for a startup trucking company's website, establishing a **three-tier architecture** (**EC2**, **S3**, **RDS**) and essential networking (**VPC**, **Security Groups**). Automated application build and deployment processes using **Jenkins CI/CD** pipelines and performed system automation via **Bash/Python** scripts.

CERTIFICATIONS

- [AWS Certified Solution Architect Associate](#)
- [Microsoft Certified Azure Administrator Associate](#)
- [Certified Kubernetes Administrator](#)

TECHNICAL SKILLS

- **Programming/Scripting Languages:** Python, SQL, Bash, Powershell
- **Automation Tools:** Jenkins, Jfrog, Nexus, Docker, Kubernetes, Ansible, Terraform
- **Containerization Tools:** Kubernetes, Docker, Docker Swarm, AKS, EKS, Openshift
- **Cloud Platforms:** AWS, Azure
- **Security Frameworks:** PCI DSS, HIPAA, ISO 2700
- **Monitoring Tools:** Splunk, CloudWatch, CloudTrail, ELK, Azure Monitor, ELK, Grafana, Prometheus

EDUCATION

Master's in cyber security | Villanova University | GPA: 3.4 | Aug 2019 – Dec 2021

Bachelor's degree in computer and network security | Wilmington University | GPA:3.6 | Jan 2016 – Dec 2019